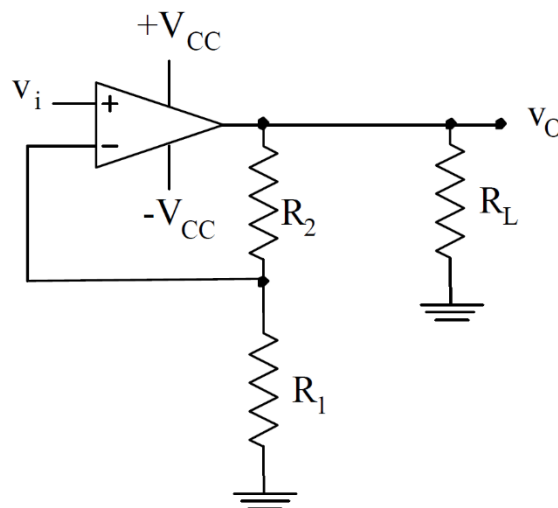


## RE2 – Stability of a feedback amplifier

Consider the circuit shown in figure where the operational amplifier (OPAMP) has an open-loop differential gain  $A_0 = 10^6$ , poles at frequencies  $\omega_{p1} = 2\pi \times 10^5 \text{ rad/s}$  and  $\omega_{p2} = 2\pi \times 10^7 \text{ rad/s}$ , infinite input resistance and zero output resistance. Consider that  $R_1 = 1\text{k}\Omega$ ,  $R_2 = 100\text{k}\Omega$  and  $R_L = 100\text{k}\Omega$ .

- Plot the Bode diagrams of the feedback gain  $A(s)\beta$ .
- Determine the frequency of the dominant pole, using compensation by displacement of the first pole, which allows the system to have a phase margin of  $45^\circ$ . Plot the Bode curves for the already compensated gain  $A(s)\beta$ . What is the resulting gain margin?
- Repeat point b) but now using compensation by insertion of additional pole.
- Compare the two compensation methods used above, stating the advantages and disadvantages of each.



## Resolution

a) and b)

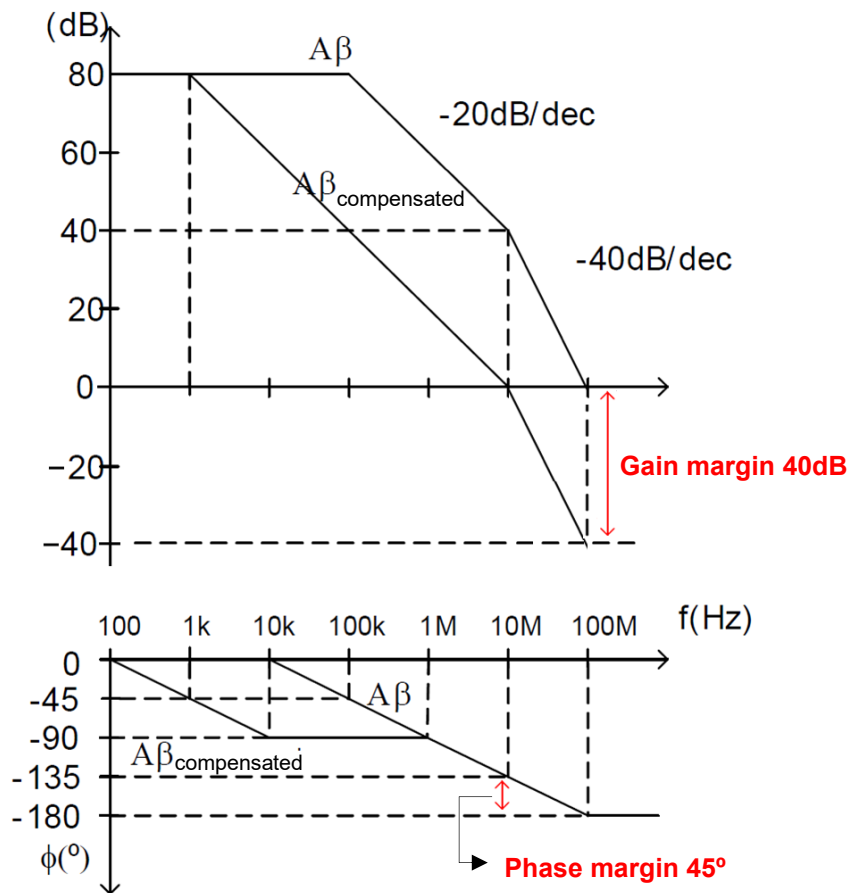
$$A_0 = 20 \log 10^6 = 120 \text{ dB}$$

$$\beta = \frac{R_1}{R_1 + R_2} \approx 0.1 \quad \Rightarrow \quad \beta = -40 \text{ dB}$$

$$A_0 \beta = 80 \text{ dB}$$

The new location of the first pole ( $f_{p1}'$ ) can be calculated graphically (see figure) or analytically:

$$f_{p1}' = \frac{f_{p2}}{A_0 \beta} = \frac{10^7}{10^4} = 1 \text{ kHz}$$

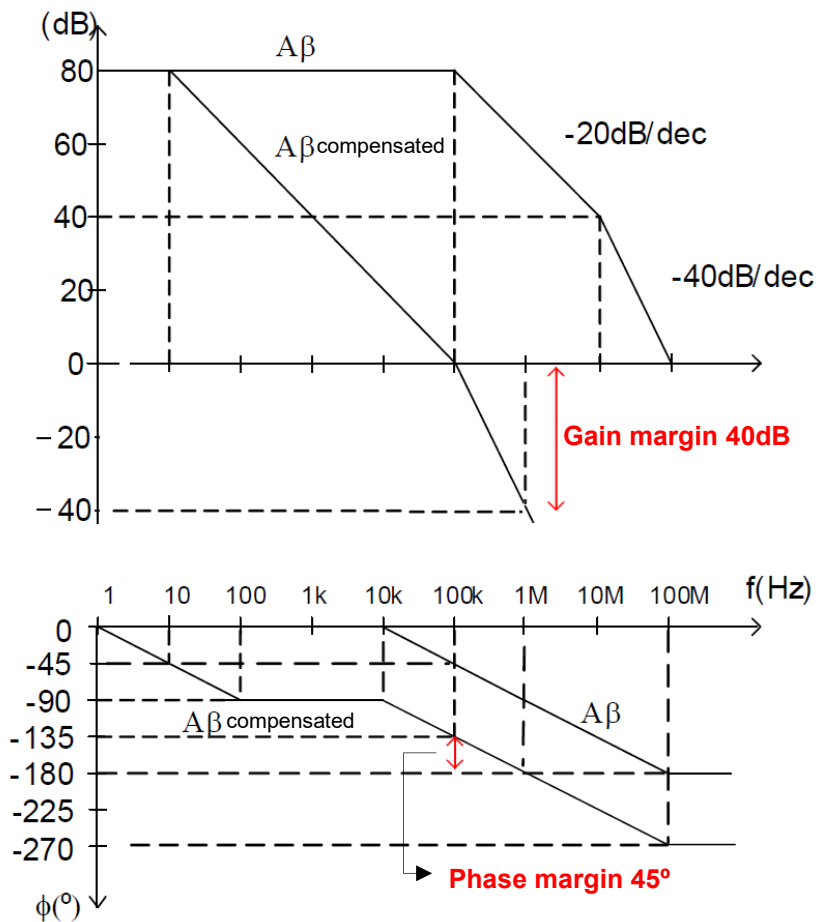


Gain margin of 40 dB because  $A\beta_{\text{compensated}} = -40 \text{ dB}$  at 100 MHz (frequency at which the phase reaches  $-180^\circ$ , asymptotically).

c)

The location of the additional pole ( $f_p$ ) can be calculated graphically (see figure) or analytically:

$$f_{p1}'' = \frac{f_{p1}}{A_0 \beta} = \frac{10^5}{10^4} = 10 \text{ Hz}$$



Gain margin of 40dB.

d)

Additional pole: Easy to implement but lower bandwidth (10Hz instead of 1kHz).

Displacement of the first pole: greater bandwidth than by additional pole but more difficult to implement.